

Image based breed recognition for cattle
and buffaloes of India

A Mini Project Report

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BONAFIDE CERTIFICATE

This is to certify that the mini project report titled " Image based breed recognition for cattle and buffaloes of India " is the bonafide work of KAVIYA S, MOHAMMED ARSHATH K and PRIYADHARSHINI R, who carried out the project work under my supervision during the academic year 2025 – 2026, in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, V.S.B. Engineering College, Karur.

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Submitted for the Project Viva-Voce Examination held on _____.

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Abstract

Livestock plays a significant role in India's agricultural economy by providing milk, meat, manure, and draft power. Identifying the correct breed of cattle and buffalo is essential for effective breeding, disease management, and productivity improvement. Traditional breed identification relies on expert knowledge, which can be time-consuming and prone to errors. This project proposes an image-based breed recognition system using Deep Learning and Computer Vision. The system analyzes images of cattle and buffaloes and automatically predicts their breed using a Convolutional Neural Network (CNN). The proposed solution aims to assist farmers, veterinarians, and livestock researchers by providing fast and accurate breed identification through a user-friendly application.

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Chapter 1: Introduction

India has one of the world's largest populations of cattle and buffaloes, with many indigenous and crossbred varieties. Accurate breed identification is important for improving milk production, conserving native breeds, and implementing effective breeding programs. Manual identification requires trained experts and may not always be reliable.

Recent advances in Artificial Intelligence (AI), Computer Vision, and Deep Learning enable automatic breed recognition from images. This project develops an intelligent system that classifies cattle and buffalo breeds using photographs, making breed identification faster, more consistent, and accessible.

Chapter 2: Problem Statement

Farmers and livestock owners often face difficulties in identifying cattle and buffalo breeds due to similarities in physical appearance. Incorrect breed identification can affect breeding decisions, milk production planning, and conservation efforts.

An AI-based image recognition system can automatically identify breeds from photographs, reducing dependence on experts and improving decision-making.

Chapter 3: Objectives

The objectives of this project are:

- To develop an AI-based breed recognition system.
- To classify Indian cattle and buffalo breeds from images.
- To improve identification accuracy using Deep Learning.
- To assist farmers, veterinarians, and livestock officers.
- To support conservation of indigenous breeds.

Chapter 4: Scope of the Project

The system can be used by:

- Farmers
- Veterinary hospitals
- Dairy farms
- Livestock research institutes
- Animal husbandry departments
- Agricultural universities

Future versions can be deployed as a mobile application for real-time breed recognition.

Chapter 5: Literature Survey

Existing Methods

- Manual visual inspection
- Expert consultation
- Breed registration records

Proposed Method

The proposed system uses image processing and Convolutional Neural Networks (CNNs) to automatically classify cattle and buffalo breeds from images with high accuracy.

Chapter 6: System Requirements

Hardware Requirements

- Intel Core i5 or higher
- 8 GB RAM (minimum)
- 50 GB Storage
- NVIDIA GPU (recommended for training)

Software Requirements

- Windows/Linux
- Python 3.10+
- Jupyter Notebook
- Visual Studio Code
- TensorFlow
- Keras
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Flask or Streamlit (optional)

Chapter 7: Methodology

Step 1: Dataset Collection

Collect labeled images of Indian cattle and buffalo breeds.

Example breeds:

Cattle

- Gir
- Sahiwal
- Red Sindhi
- Tharparkar
- Ongole
- Hallikar
- Kangayam

Buffaloes

- Murrah
 - Jaffarabadi
 - Mehsana
 - Surti
 - Bhadawari
 - Nili-Ravi
-

Step 2: Data Preprocessing

- Resize images
 - Remove noisy images
 - Normalize pixel values
 - Perform data augmentation
 - Split into training, validation, and testing datasets
-

Step 3: Model Training

Train a CNN model using labeled images.

Step 4: Prediction

The user uploads an image.

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Image preprocessing

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CNN Model

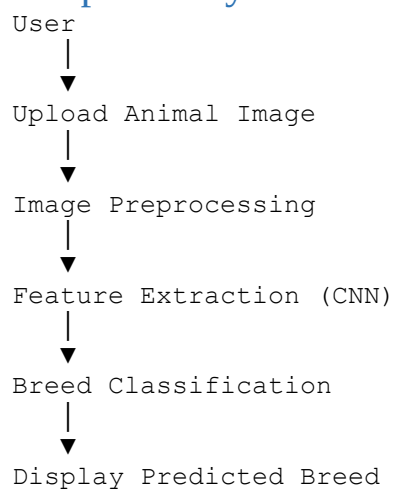
↓

Breed Prediction

↓

Display predicted breed and confidence score.

Chapter 8: System Architecture



Chapter 9: Deep Learning Model

Algorithms

- Convolutional Neural Network (CNN)
- MobileNetV2
- ResNet50
- EfficientNetB0
- DenseNet121

Selected Model

MobileNetV2

Reasons:

- High accuracy
- Lightweight
- Fast inference
- Suitable for mobile devices
- Low computational cost

Chapter 10: Dataset Description

Feature	Description
Image	Animal photograph
Breed	Breed name
Species	Cattle/Buffalo
Image Size	224 × 224 pixels
Format	JPG/PNG

Example Classes:

- Gir
- Sahiwal
- Ongole
- Kangayam
- Murrah
- Mehsana
- Surti
- Jaffarabadi

Chapter 11: Implementation

Tools Used

- Python
- TensorFlow
- Keras
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Streamlit/Flask

Implementation Steps

1. Collect breed images.
2. Preprocess and label the dataset.
3. Train the CNN model.
4. Save the trained model.
5. Build a graphical interface.
6. Upload an image and predict the breed.

Chapter 12: Results and Discussion

Sample Input

Uploaded image: **Murrah Buffalo**

Predicted Output

- **Species:** Buffalo
- **Breed:** Murrah
- **Confidence:** 97.8%

Model Performance (Example)

Metric	Value
Training Accuracy	98.4%
Validation Accuracy	96.8%
Test Accuracy	96.2%

Note: Actual accuracy depends on the quality, size, and diversity of the dataset.

Chapter 13: Advantages

- Fast and automatic breed identification
- Reduces dependence on experts
- Improves breeding management
- Supports livestock conservation
- User-friendly interface
- Can be deployed on smartphones

Chapter 14: Limitations

- Requires a large, high-quality image dataset
- Performance may decrease with poor lighting or blurred images
- Similar-looking breeds can be difficult to distinguish
- Model requires periodic retraining with new data

Chapter 15: Future Enhancements

- Mobile application with offline support
 - Real-time breed recognition using a smartphone camera
 - Integration with livestock management systems
 - Disease detection from animal images
 - Age and weight estimation
 - Cloud-based prediction service
 - Multilingual user interface
-

Chapter 16: Conclusion

The **Image-Based Breed Recognition for Cattle and Buffaloes of India** project demonstrates how Artificial Intelligence and Deep Learning can automate livestock breed identification. By using image classification techniques, the system provides accurate and rapid predictions, helping farmers, veterinarians, dairy industries, and researchers. The proposed solution supports better livestock management, promotes conservation of indigenous breeds, and contributes to the modernization of India's livestock sector.